

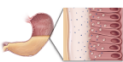
Researchers are interested in studying gene expression profiles of connective tissue. Given this, they should analyze gene expression for all of the following tissues EXCEPT:

- ☐ A. bone tissue. [8%]
- ☐ B. blood tissue. [19%]
- ☐ C. adipose tissue. [35%]
- ✓ ☒ D. muscle tissue. [36%]

Correct

37% answered correctly

Explanation:

Types of tissue				
	Epithelial tissue	Muscle tissue	Connective tissue	Nervous tissue
Body tissue type				
Germ layer precursor	Endoderm Mesoderm Ectoderm	Mesoderm	Mesoderm	Ectoderm
Functions	• Lines surfaces • Protection • Absorption, filtration, and secretion	• Voluntary movement (skeletal muscle) • Involuntary movement (cardiac and smooth muscle)	• Support and protection • Attachment of tissues • Insulation and thermoregulation • Transport	• Transmission and receipt of electrical signals • Response to internal and external stimuli (eg, sensory receptors)
Examples	• Skin • Lining of hollow organs (eg, digestive tract and trachea)	• Skeletal (attached to bone) • Cardiac (heart) • Smooth muscle (hollow organs)	• Bone • Blood • Fat (adipose) • Tendons/ligaments • Cartilage	• Brain • Spinal cord • Nerves

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There are **four main tissue types** in the human body: epithelial, muscle, connective, and nervous tissue. Connective tissue, consisting of specialized cells that secrete and maintain an extensive extracellular matrix, has several functions. Connective tissue serves to attach tissues and organs to one another and is essential for the protection and support of surrounding structures.

In the given scenario, researchers are interested in studying gene expression profiles of connective tissue. Therefore, they should analyze gene expression for several **subtypes of connective tissue**, including:

- **Bone**, which makes up the **skeletal system** and provides structural support and physical protection for internal organs as well as a framework for mobility. Bone comprises osteoblasts (matrix-secreting cells that build bone), osteoclasts (bone-degrading cells), and osteocytes (mature bone cells) (**Choice A**).
- **Blood**, which transports gases, cells (ie, erythrocytes, leukocytes), nutrients, wastes, and other materials (eg, hormones) throughout the body via blood vessels. Although mature erythrocytes have no nucleus, gene expression profiles can still be obtained from leukocytes (**Choice B**).
- Adipose tissue, which is responsible for the storage of fats as well as providing

cushioning for organs and insulation to prevent heat loss (**Choice C**).

Muscle tissue, classified into **three types**, is under the control of the nervous system and allows for movement. Although connective tissue comes into contact with **muscle tissue**, muscle is *not* considered to be a type of connective tissue. Therefore, researchers interested in studying gene expression in connective tissue samples should *not* include muscle tissues in their analysis.

Educational objective:

The body consists of four types of tissue: epithelial, muscle, connective, and nervous tissue. Connective tissue provides protection and support for organs. Types of connective tissue include bone, blood, and adipose tissue.

Time spent:0 sec

Subject:
Biology

QID:402033

System:
Musculoskeletal System